Environmental Science and Pollution Research (2021) 28:58677–58687
https://doi.org/10.1007/s11356-021-14810-9

RESEARCH ARTICLE

Pilot-scale microaerobic hydrolysis-acidification and anoxic-oxic processes for the treatment of petrochemical wastewater

Libing Chu[1] · Pengyuan Ding[2] · Mingcong Ding[3]

Received: 8 January 2021 / Accepted: 7 June 2021 / Published online: 12 June 2021
© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2021

Abstract

Microaerobic hydrolysis and acidification (MHA), as a promising pre-treatment method of industrial wastewater, is drawing increasing attention to enhance the hydrolysis-acidification rate and inhibit the production of toxic gas H_2S. In the present work, a pilot-scale MHA reactor coupled with anoxic-oxic (A/O) processes for treating the petrochemical wastewater was established and the mechanism and application of the MHA reaction were explored. The results showed that the ratio of VFA/COD was increased by 43–90% and low effluent S^{2-} concentration (less than 0.2 mg/L) was obtained after MHA treatment with 5.5–13.8 L air m^{-3} h^{-1} supply. The MHA sludge exhibited a good settleability, a higher protease activity and plentiful community diversity. In addition to the dominant anaerobic bacteria responsible for hydrolysis and acidification such as *Clostridiales* uncultured, *Anaerovorax*, *Anaerolineaceae* uncultured and *Fastidiosipila*, the sulfate reducing bacteria involving *Desulfobacter*, *Desulfomicrobium* and *Desulfobulbus*, the sulphur oxidizing bacteria involving *Thiobacillus*, *Arcobacter* and *Limnobacter*, the nitrifies such as *Nitrosomonadaceae* uncultured and *Nitrospira*, and denitrifies *Thauera* were also identified. MHA pre-treatment guaranteed the efficacy and stability of the following A/O treatment. The removal efficiency of COD and ammonium of the MHA-A/O system remained at around 78.3% and 80.8%, respectively, although the organic load fluctuated greatly in the influent.

Keywords Microaeration · Biodegradation · Enzyme activity · Wastewater treatment · Bioreactors · Illumina sequencing

Introduction

The treatment of petrochemical wastewater produced from petroleum refinery and manufacture of organic chemical materials such as ethylene, toluene, fiber, resins, plastics, and pesticides, has become a challenging issue (Diya'uddeen et al. 2011). Petrochemical wastewater contains many recalcitrant hazardous pollutants involving aromatic hydrocarbons, phenolics, heterocyclic compounds, aniline, etc. (Rubio-Clemente et al. 2015). Therefore, the processes combined of pre-treatment and secondary biological treatment are often adopted to treat the refractory industrial wastewater (Guo et al. 2014). The pre-treatment methods available include coagulation, chemical oxidation and hydrolysis, and acidification (HA), etc. (Kakavandi and Babaei 2016; Lin et al. 2001; Rubio-Clemente et al. 2015). Among them, HA pre-treatment is promising owing to the fact that it was effective to improve the biodegradability of wastewater. In addition, HA process was economic and can withdraw the shock load (Ding et al. 2016; Wu et al. 2016).

Hydrolysis and acidification process is generally operated at anaerobic conditions. Recently, studies have been more concerned about the implementation of microaeration in the anaerobic digestion system (Chen et al. 2020; JGdS et al. 2021; Loughrin et al. 2020; Ramos and Fdz-Polanco 2014; Ruan et al. 2019). Microaeration hydrolysis and acidification (MHA) is to dose small amounts of air into the HA reactor. It was reported that the microaerobic condition contributed to enhance the acidification degree, improve the biodegradability

Responsible Editor: Robert Duran

✉ Libing Chu
chulibing@tsinghua.edu.cn

[1] Institute of Nuclear and New Energy Technology, Tsinghua University, Beijing 100084, People's Republic of China

[2] Petroleum Exploration and Production Research Institute, Sinopec, Beijing 100083, People's Republic of China

[3] College of Electrical and Information Engineering, Beijing University of Civil Engineering and Architecture, Beijing 102616, People's Republic of China

of organic pollutants, and reduce H_2S production in the biogas, leading to odor-free effluent (Jenicek et al. 2008; Krayzelova et al. 2015; Wu et al. 2018).

Lim et al. (2014) reported that with 5–7 mL O_2 L^{-1} d^{-1} supply in the HA reactor treating brown water and food waste, the levels of soluble COD and volatile fatty acids (VFA) increased by 43.1–32.0% and 2.5–3.9%, respectively compared to that under the anaerobic conditions. Chen et al. (2008) found that the BOD_5/COD ratio of pharmaceutical wastewater increased from initial 0.20 to 0.36 by HA process under the microaerobic condition, which was higher than that under the anaerobic condition (0.30). Diaz et al. (2011) found that the amount of H_2S in the outflow biogas decreased dramatically from 14.4 L m^{-3} by anaerobic digestion to below 0.3 L m^{-3} at the microaeration conditions with 1.8 mL O_2 min^{-1} supply and elemental sulfur was the final product. Our previous lab-scale studies showed that MHA was efficient to pre-treat the mixture of petrochemical wastewater and domestic wastewater (Yang et al. 2015). The efficiency of COD removal was 33–42% in the MHA reactor and the ratio of BOD_5/COD in the MHA effluent reached 0.34.

In the present work, the pilot-scale MHA reactor was developed to pre-treat petrochemical wastewater. Studies were focused on the microbial diversities and activities of sludge in MHA reactor which were rarely considered in previous studies. In addition, the effect of microaeration intensity on acidogenic degree and S^{2-} production in MHA reactor was assessed. The performance of MHA pre-treatment coupled with anoxic/oxic (A/O) biological treatment processes was evaluated in terms of organics and nutrients removal, composition of organic pollutants, and changes in the sludge morphology. Taken together, this research would provide the useful information about the mechanism of the MHA reaction and contribute to the practical application of MHA-A/O system in the treatment of industrial wastewater.

Materials and methods

Pilot-scale set-up

The pilot-scale MHA-A/O set-up was made of carbon steel coated with fluoroplastic for corrosion protection. It was built in a test workshop of a Petrochemical Wastewater Treatment Plant (PWTP) located in a chemical industrial park of northeast China. Figure 1 presents the diagram and picture of the experiments set-up. The working volume of MHA and A/O reactors was 14.5 m^3 and 20.3 m^3, respectively. The A/O reactor is composed of 6 compartments, one served as anoxic tank (A) and the other 5 compartment served as oxic tank (O1–O5). There is an agitator in the anoxic tank to mix sludge and wastewater. Air was introduced into the MHA reactor and the oxic tanks through an air compressor and the flowrate was controlled at 0.08–0.2 m^3/h in the MHA tank and 8.0 m^3/h in the oxic tank by rotameters, respectively.

The raw petrochemical wastewater was settled in the primary setting tank and the supernatant was pumped into the MHA reactor through a distribution reservoir. The effluent from MHA tank after settlement entered into the A/O reactor and the sludge settled was returned into the MHA reactor. The effluent from A/O reactor entered into the secondary clarifier. The supernatant was discharged into the sewer and sludge was returned into the anoxic reactor with internal recycle ratio of 100%. A certain amount of excess sludge was withdrawn from the secondary clarifier once a day to keep the sludge retention time of 17–24 days.

Source and characteristics of petrochemical wastewater

The raw petrochemical wastewater was derived from more than 70 chemicals production plants in the chemical industrial park. A variety of chemicals were made in the park including ethylene, acrylate, isobutene, epoxy ethane, acrylonitrile butadiene styrene, vanillic aldehyde, fertilizers, pesticides, resins, dyes, and refined oil, etc. Table 1 shows the characteristics of the influent of MHA-A/O reactor. The influent COD and ammonium fluctuated greatly which is typical for the actual industrial wastewater. The soluble COD accounted for 89.2% in the total COD and a high level of sulfate were observed.

Operational procedure of the pilot-scale MHA-A/O system

The MHA and A/O reactors were inoculated with the return sludge of the PWTP. The influent load was increased gradually by increasing the feeding rate according to the performance of the bioreactors. When the COD removal reached more than 15% in the MHA reactor and 70% in the A/O reactor, the flowrate of influent was increased. The MHA-A/O bioreactors were started up successfully after a month of operation. Thereafter, the hydrolytic retention time was controlled at 14 h and 20 h in the MHA and A/O reactor, respectively. The levels of dissolved oxygen (DO) in MHA, anoxic and oxic tank remained at 0.15–0.32 mg/L, 0.12–0.28 mg/L, and 2.8–5.9 mg/L, respectively. The sludge retention time was 17–24 days. The oxidation reduction potential (ORP) ranged from −240 to −360 mV in the MHA tank and from −102 to −215 mV in the anoxic tank, respectively. The temperature remained at 25–27 °C. The pH values in the influent and the effluent from MHA, A and O tanks were 8.18, 7.74, 7.44, and 7.23, respectively. The inlet air flowrate in MHA tank was controlled at 0.08–0.1 m^3/h during the first 5 months of operation and increased to 0.2 m^3/h in the last month to study the impact of aeration intensity.

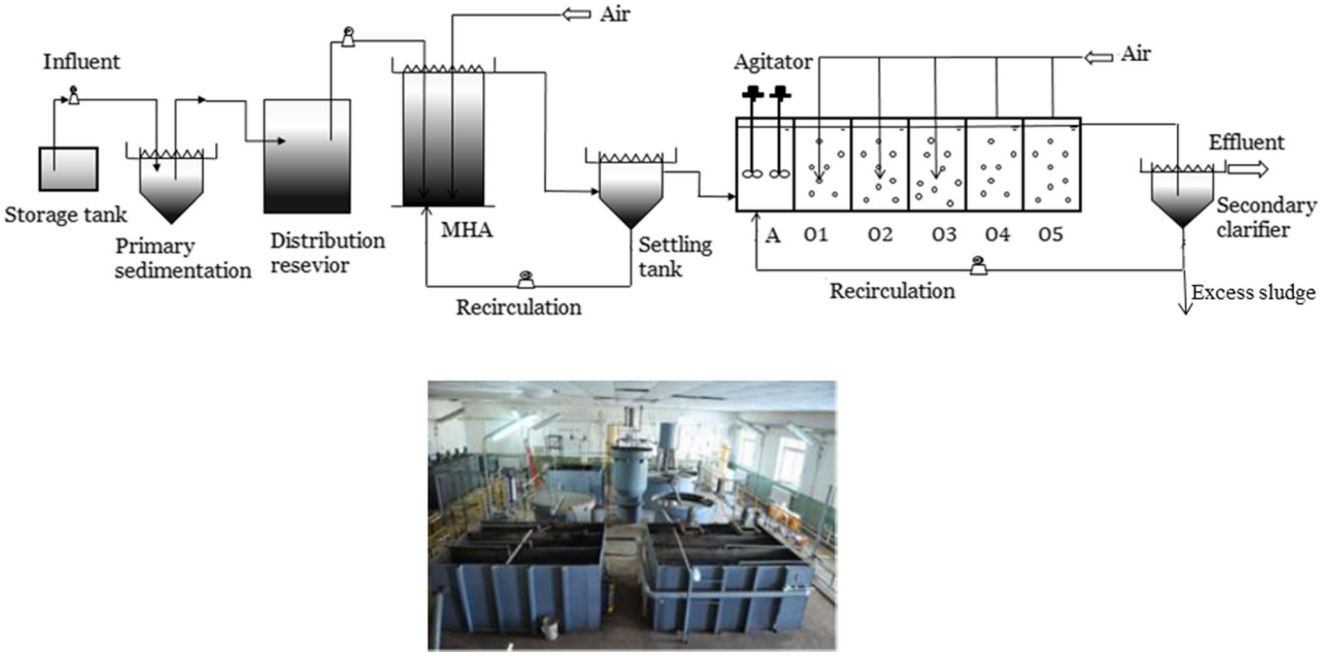


Fig. 1 The schematic diagram of the pilot-scale set-up and picture of worksite

The samples of influent and effluent from MHA and secondary clarifier were taken periodically to detect the content of COD, UV_{254}, total nitrogen (TN), ammonium (NH_4-N), nitrate (NO_3-N), nitrite (NO_2-N) and total phosphorus (TP), etc. The values of total suspended solids (TSS), volatile suspended solids (VSS), SV (%) and SVI (mL g^{-1}) of sludge in the three tanks were monitored. For MHA reactor, the concentration of VFA, sulfate (SO_4^{2-}) and sulfite (S^{2-}) in the influent and effluent was evaluated. The distribution of COD, NH_4-N, UV_{254}, and NO_3-N levels along the different parts of the MHA-A/O bioreactors was evaluated at monthly interval and the representative data were shown in the text. All water samples were centrifuged at 9838 g for 5 min and the supernatant obtained was collected for analysis.

When the operation of bioreactor reached steady state, the composition of organic pollutants in the influent, MHA effluent and final effluent was assessed by GC-MS. The sludge samples in the three tanks were taken to analyze the microbial population and activity involving protease and dehydrogenase activity. The microbial activity of biomass was closely related to the substrates degradation rate. The activity of hydrolases such as protease enables to reflect the microbial activity to degrade the organic substrates (Gessesse et al. 2003). Biological oxidation of the organic substances is a dehydrogenation process and all living microorganisms have a dehydrogenase activity, which is considered to be a sensitive method for determining microbial activity (Lazarova and Manem 1995).

Analytical methods

COD, nitrogen compounds, TP, sludge properties (TSS, VSS, SV, and SVI) were evaluated according to the Standard Methods (APHA/AWWA/AWEF 2017). VFA was detected using titration method. UV_{254} was measured by a Lambda25 UV-Vis spectrophotometer (PerkinElmer, USA). pH was detected using a pH meter (SX721, Shanxin, China). ORP and DO concentrations were evaluated using a portable WTW DO/ORP meter (multi 3420, Germany).

The identification of the organic pollutants was conducted using Agilent GC-MS analyzer (7890A/5975C, USA) equipped with OV-101 capillary column. Before GC-MS analysis, the organic pollutants were extracted from the wastewater samples using dichloromethane as the extraction reagent. The temperature of GC injection port was 280 °C and the carrier gas was helium. The oven temperature program was as follows: initializing to 55 °C for 4 min and raising to

Table 1 The properties of the raw petrochemical wastewater

COD (mg/L)		UV_{254}	N compounds (mg/L)				TP (mg/L)	SO_4^{2-} (mg/L)	pH
COD_t	COD_{dis}		TN	NH_4-N	NO_3-N	NO_2-N			
517.2 ± 112.1	461.1 ± 66.5	2.2 ± 0.8	33.1 ± 3.0	28.5 ± 8.3	4.1 ± 2.2	0.06 ± 0.03	1.2 ± 0.8	732.4 ± 209.3	8.2 ± 0.7

280 °C with a step of 10 °C min^{-1}. The ion source temperature of MS detector was 200 °C.

The protease activity of sludge (U/g VSS) was determined using 0.5% azocasein as chromogenic substrate (Kim et al. 2002). One unit of protease activity (1 U) represents to degrade 1 mg of azocasein at 37°C and reaction of 90 min. The dehydrogenases activity of sludge was determined using triphenyl tetrazolium chloride (TTC) as indicator (Tarjányi-Szikora et al. 2013). TTC was reduced into red-colored triphenyl formazan (TF) which can be quantified by spectrometry. The dehydrogenases activity was expressed as mg TF/g VSS.

The microbial community of sludge in different tanks was analyzed by Illumina MiSeq sequencing (Yang et al. 2015). DNA was extracted using OMEGA soil DNA extraction kit (E.Z.N.A., USA) and examined using 1% agarose gel electrophoresis. The specific primers 338F (ACTCCTACGGGAGG CAGCA) and 806R (GGACTACHVGGGTWTCTAAT) with barcode was designed and synthesized. The PCR reaction was carried out in solution of 20 μL involving 5×FastPfu Buffer 4 μL, forward primer 0.8 μL, reverse primer 0.8 μL, template DNA 10 ng, dNTPs 2 μL, FastPfu Polymerase 0.4 μL and ddH_2O. Thermal cycling included 1× (3 min at 95°C), 27× (30 s at 95°C, 30 s at 55 °C, 45 s at 72°C), 1× (10 min at 72°C) and 10°C until halted by user. Then, the PCR products were examined by 2% agarose gel electrophoresis, purified using the AxyPrep DNA Gel Extraction Kit according to manufacturer's protocols and quantified using Quantus™ Fluorometer (Promega, USA). Finally, the purified amplicons were pooled in equimolar and paired-end sequenced on an Illumina MiSeq PE300 platform according to the standard protocols by Majorbio Bio-Pharm Technology Co. Ltd. (Shanghai, China). The raw 16S rRNA gene sequencing reads were demultiplexed, quality-filtered by fastp version 0.20.0 and merged by FLASH version 1.2.7. The average length of the bases was 441 bp. The clustering and taxonomy analysis of Operational Taxonomic Units (OTUs) was conducted at a similarity level of 97% using UPARSE version 7.1. Chimeric sequences were identified and removed. The taxonomy of each OTU representative sequence was analyzed by RDP Classifier version 2.2 against the 16S rRNA database (e.g. Silva v138) using confidence threshold of 0.7.

Results and discussion

Organics and nutrients removal

After start-up, the pilot-scale MHA-A/O bioreactor was operated continually for 6 months, except for the period from 22nd to 30th day when the feeding of the bioreactor was stopped owing to the shut-down of the PWTP for maintenance. Figure 2 depicts the variation of COD and NH_4-N concentrations in influent and effluent as well as the efficiency of COD and NH_4-N removal. Although COD concentration in the influent fluctuated greatly, it remained at 98.9 ± 27.5 mg/L in the final effluent. The COD removal of the overall system ranged 70–87% with the average of 78.3%, while around 20% of COD reduction was achieved in MHA reactor. Similar results were reported by other researchers. Song et al. (2019) studied the treatment of 2-butenal manufacture wastewater by MHA process. Results showed that around 23.7% of COD removal was obtained and the acidification degree could reach 20.9%.

The efficiency of NH_4-N removal was higher than that of COD removal except for some periods when the influent NH_4-N concentration was much high (up to 80 mg/L). The efficiency of NH_4-N removal reached 80.8 ± 15.7% and a low level of 5.9 ± 2.9 mg NH_4-N/L was observed in the final effluent. This demonstrates that the activity of nitrifying bacteria was high in the A/O reactor. The NH_4-N concentration increased after MHA treatment owing to ammonification.

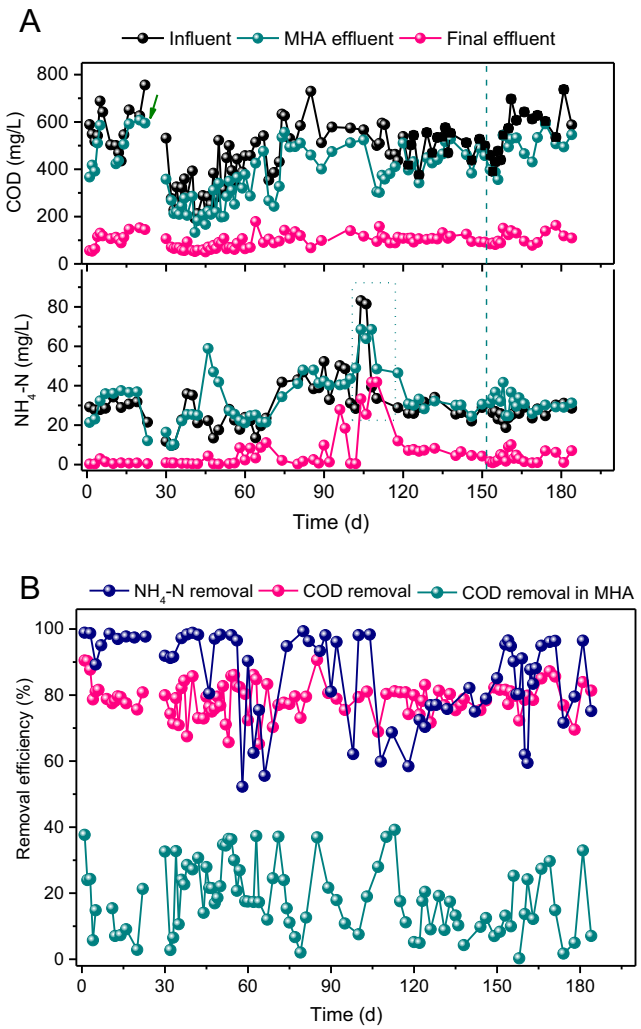


Fig. 2 Changes in COD and NH_4-N concentrations in the influent and effluent (**A**) and their removal efficiencies (**B**)

This could also suggest that the activity of ammonifying bacteria was not affected under the microaerobic conditions, leading to the conversation of organic nitrogen to ammonium. Some nitrogen-containing organic pollutants such as pyridine, aminobenzene and acetochlor etc. were identified in the raw petrochemical wastewater according to GC-MS results (as shown in the following Fig. 4). The effluent TN concentration maintained at 16.9 ± 5.0 mg/L and the TN removal was around 47.3%. The TP level in the effluent was 0.43 ± 0.47 mg/L and around 64.3% of TP removal was achieved by MHA-A/O treatment.

Figure 3 presents the alteration of COD, NH_4-N, UV_{254} and NO_3-N levels along the different part from inlet to outlet of the system. The NO_2-N concentration in the effluent was very low (0.04 ± 0.02 mg/L). This suggested that nitrification in A/O reactor took place completely. As shown in Fig. 3, each part of the system contributed to the removal of COD and UV_{254}, while the degradation occurred mainly in the A/O reactor. The absorbance of UV_{254} was eliminated by 18% after MHA treatment and more than 60% after A/O treatment. This suggests that the aromatic compounds having C=C and C=O double bonds were removed mainly in A/O reactors. After MHA pre-treatment, the macro-molecular and recalcitrant pollutants are converted into the micro-molecular and less toxic substances which were then transformed into the volatile organic acid such as acetate and butyric acid (Wang et al. 2014). Those substances could be degraded easily into CO_2 and water by the following A/O biological treatment.

The performance of the pilot-scale MHA-A/O system was comparable to that of the bench-scale bioreactor reported in our previous studies. The effluent COD values were similar and NH_4-N concentration was lower in the bench-scale system. Around 99 mg COD/L and 1.3 mg NH_4-N/L were obtained in the effluent of the bench-scale MHA-A/O bioreactors (Ding et al. 2016; Yang et al. 2015). It should be noted that the feeding wastewater of the bench-scale system was composed of petrochemical wastewater and domestic wastewater (3:1). In the present pilot-scale MHA-A/O system, the wastewater was totally from the petrochemical industry. The Environmental Protection Agency of China has published a new strict emission standard of pollutants in petrochemical wastewater (GB 31571-2015). The permissible discharge level of COD declined from 100 mg/L in the old standard (GB8978-1996) to 60 mg/L. The new standard for discharging NH_4-N, TN, and TP was 8.0 mg/L, 40 mg/L, and 1.0 mg /L, respectively. The results showed that the NH_4-N, TN, and TP concentrations in the effluent from the pilot-scale MHA-A/O bioreactor have met the new discharge standard, but the COD content was higher. The advanced treatment processes after MHA-A/O bioreactor seem to be necessary to attain COD limit.

Effect of microaeration intensity

Table 2 shows effects of the microaeration intensity on the acidification degree and H_2S production in MHA reactor. The efficiency of acidogenic fermentation was high within the microaeration intensity applied. The ratio of VFA/COD in the MHA effluent was increased by 43% with the aeration intensity of 5.5–6.9 L air m^{-3} h^{-1} and by 90% with 13.8 L air

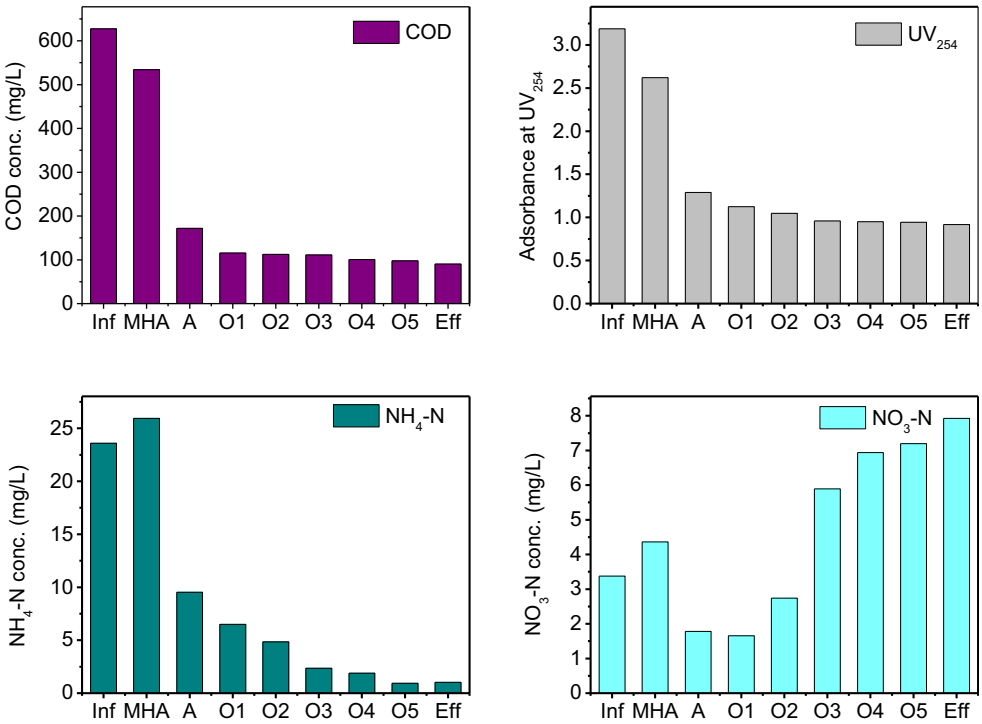


Fig. 3 Variation of COD, NH_4-N, UV_{254} and NO_3-N levels along the different part of the MHA-A/O system

Table 2 Effect of microaeration intensity on acidification and S^{2-} production in MHA reactor

Aeration (L air/m^3 h)	VFA (mmol/L)		VFA/COD (mg/mg)		S^{2-} (mg/L)		SO_4^{2-} (mg/L)	
	Influent	Effluent	Influent	Effluent	Influent	Effluent	Influent	Effluent
5.5–6.9	2.37 ± 0.89	2.68 ± 1.00	0.35 ± 0.25	0.40 ± 0.16	0.08 ± 0.05	0.22 ± 0.13	702.2 ± 240.3	703.0 ± 195.9
13.8	1.86 ± 0.57	3.07 ± 0.43	0.20 ± 0.06	0.38 ± 0.09	0.08 ± 0.03	0.19 ± 0.16	802.8 ± 80.5	796.5 ± 83.9

m^{-3} h^{-1}, respectively compared to that in the influent. The SO_4^{2-} concentrations were similar in both the influent and effluent. The S^{2-} concentration in the MHA effluent decreased with increasing the aeration intensity and it maintained at around 0.2 mg/L, which confirmed the advantage of MHA in the reduction of odors gas H_2S.

Characterization of organic pollutants

As shown in Fig. 4, a variety of organic compounds including methacrylate, pyridine, chlorobenzene, styrene, aminobenzene, acetophenone, naphthalene, acetochlor, etc. were identified in the raw petrochemical wastewater. Those compounds are the raw materials/intermediates/monomers to produce plasticizer, dyes, resin, pesticide, etc. After MHA treatment, almost a half of pollutants disappeared, while only two compounds were detected and their abundance was declined remarkably after A/O treatment. This indicates the effectiveness of MHA-A/O treatment in decomposition of the toxic organic pollutants. In the new standard (GB 31571-2015), the discharge limits to more than 60 pollutants involving bisphenol A (0.1 mg/L), polycyclic aromatic hydrocarbon (0.02 mg/L), acrylonitrile (2.0 mg/L), styrene (0.2 mg/L) and aminobenzene (0.5 mg/L) were required. The results of GC-MS analysis demonstrated that the MHA-A/O processes were promising to treat petrochemical wastewater to meet the new requirement in discharging the organic pollutants.

Alternation of sludge concentration and property

Figure 5 reveals the changes in sludge concentration, the ratio of VSS/TSS, SV and SVI indices in MHA, A and O tanks during the half a year of operation. The TSS concentrations in the MHA tank fluctuated in the ranges of 2.8–6.5 g/L, which might be attributed to the effect of the microaeration on sludge retention. The microaeration operation should be controlled carefully to prevent the sludge washout. The ratio of VSS/TSS of MHA sludge decreased from 64% to 48%, which indicated that the accumulation of the inorganic substances occurred in the MHA tank. However, the pre-treatment of MHA guaranteed the efficacy of A/O reactor. The sludge concentration increased gradually in the anoxic and oxic tanks and ratio of VSS/TSS remained at a high level of around 70%, which is consisted with the high degradation efficiency of A/O reactor. It is noted that the introduction of limited oxygen did not affect the settleability of sludge. SV and SVI of sludge in the three tanks maintained at relatively low levels

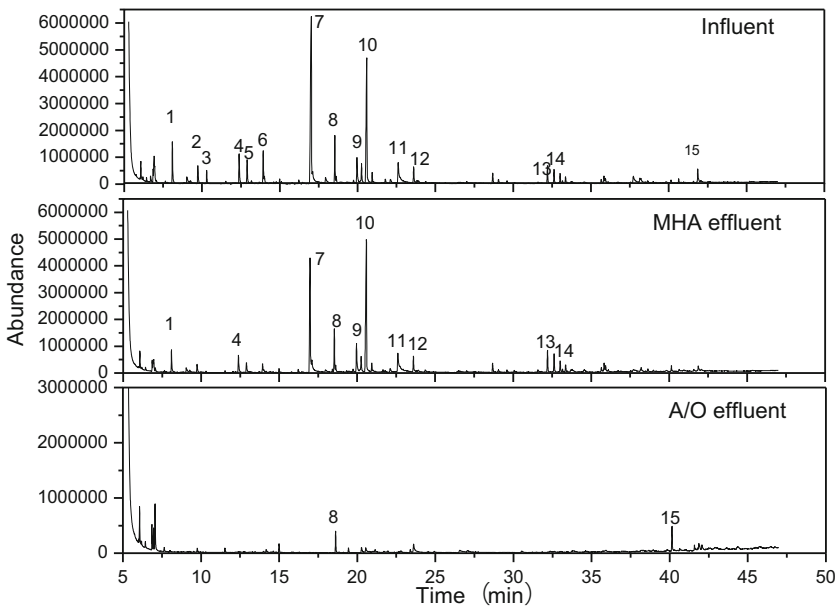


Fig. 4 GC-MS spectrum of influent, MHA effluent and A/O effluent (1 methyl methacrylate; 2 pyridine; 3 2,3-dichloro-2-methyl butane; 4 chlorobenzene; 5 styrene; 6 4-methylcyclohexanone; 7 aminobenzene 8 4-methayl succinonitrile; 9 methylbenzylalcohol; 10 acetophenone; 11 methylbenzyl alcohol; 12 naphthalene; 13 phenethylamine methyl; 14 2-phenyl-N-acetamide; 15 acetochlor)

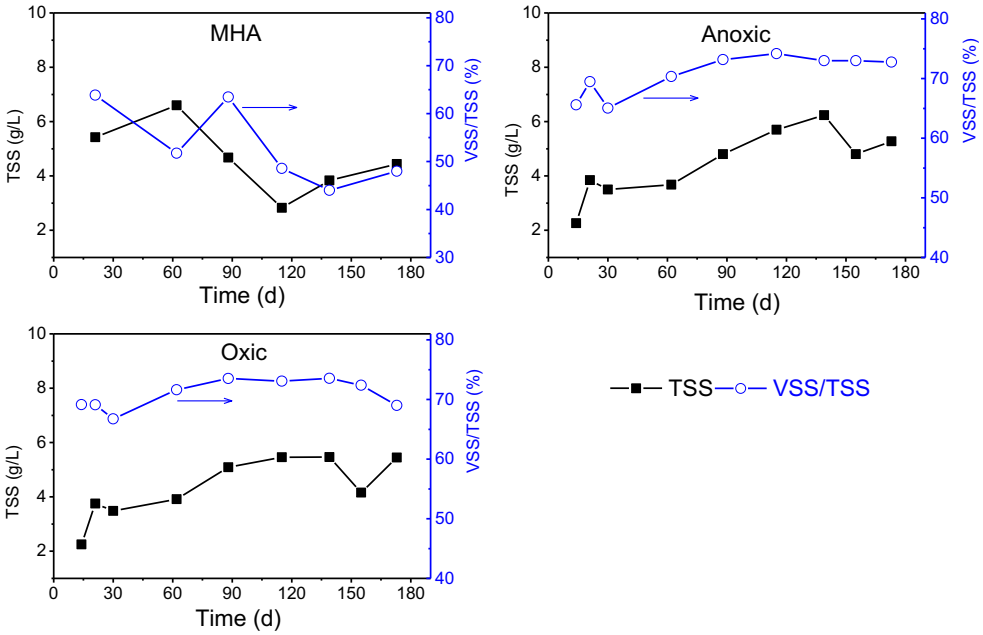


Fig. 5 Changes in sludge concentration and VSS/TSS ratio in MHA, A and O tank

(around 20% and 60 mL/g) for the floc sludge, which demonstrated the good settling property.

Microbial activity and community

Figure 6 presents the activities of protease and dehydrogenases of the sludge in MHA, anoxic and oxic tanks. The MHA sludge exhibited a higher protease activity than the A/O sludge, whereas the dehydrogenases activity was higher for the sludge in the A and O tanks. The results of enzymes activity were consistent with the function in each tank. The higher protease activity of the biomass facilitated to improve the hydrolysis rate of the macro-molecular substances in the petrochemical wastewater. The organic pollutants were oxidized into CO_2 and water mainly by the A/O treatment corresponding to the higher dehydrogenase activity.

The Illumina MiSeq sequencing obtained 22,375 sequences for each sludge sample with high Good's coverage of more than 99%. The MHA sludge had the highest community richness and diversity with Ace and Simpson indices of 941 and 0.0156, following by the oxic sludge (Ace index 906 and Simpson index 0.0153) and anoxic sludge (Ace index 899 and Simpson index 0.0304). Figure 7 reveals the taxonomic composition of sludge in the three tanks. The sludge in the A and O tank has a higher similarity. At the phylum levels, *Proteobacteria* and *Firmicutes* were predominant in MHA sludge, while *Proteobacteria* and *Bacteroidetes* were highly enriched in A and O sludge. Similar results were observed by other studies. Lim et al. reported that more diverse bacteria community was detected in the MHA reactor (18 OTUs affiliated to 4 phyla), whereas only 7 bacterial OTUs (affiliated to 2 phyla) were identified in the anaerobic reactor (2014). Bacteria in phylum *Firmicutes* with the function of metabolizing protein, lipids, cellulose and other macro-molecular

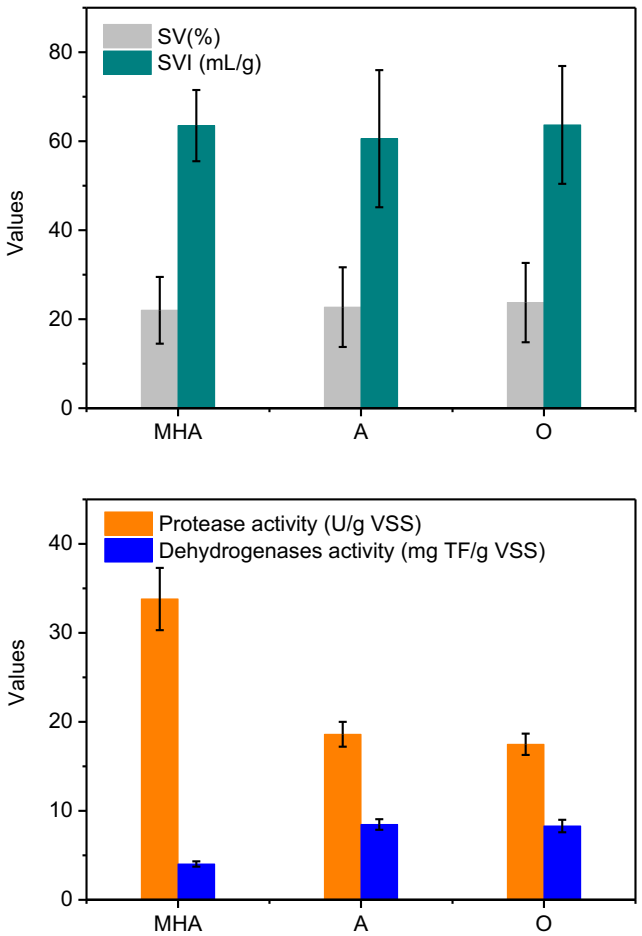


Fig. 6 Variation of activities and settleability of sludge in MHA, A and O tank

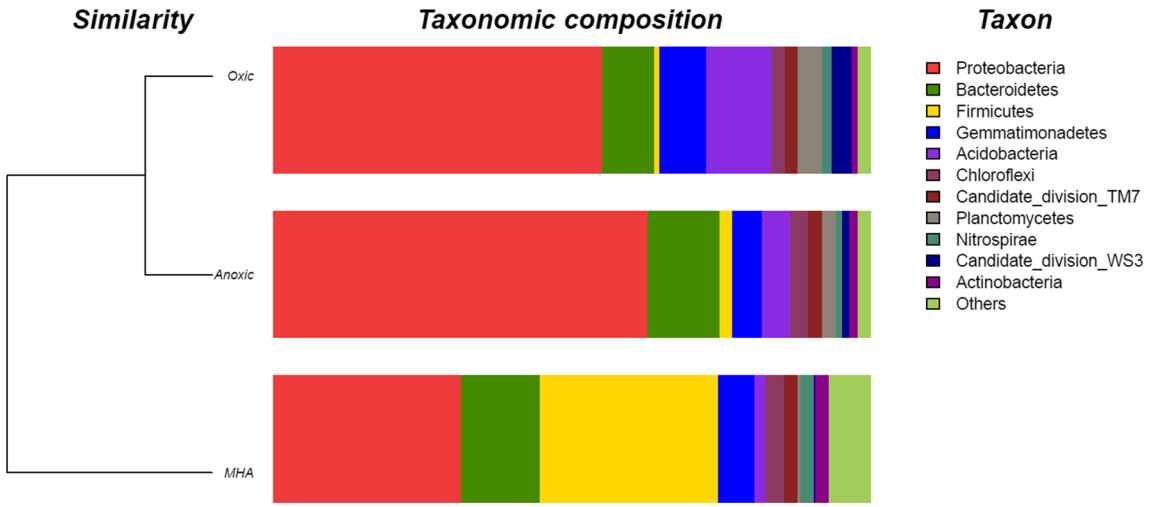


Fig. 7 The taxonomic composition of sludge in the three tanks

organics were more abundant in MHA reactor, leading to a higher hydrolysis efficiency. *Proteobacteria* is the largest phylum of bacteria and includes broad variety of members of degrading organic pollutants degradation and denitrification (Snaidr et al. 1997). *Bacteroidetes* are commonly identified in fermentation and acidification bioreactors serving to decompose the macromolecular substances such as cellulose, fiber, starch, and protein (Nakasaki et al. 2009).

The heatmap generated from the major genera in sludge samples was shown in Fig. 8. Bacteria belonging to *Clostridiales* uncultured, *Christensenellaceae* uncultured, *Gemmatimonadaceae* uncultured, *Thauera*, *Fastidiosipila*, *Betaproteobacteria* unclassified and *Anaerovorax* were dominant in MHA bioreactor. The members enriched highly in A/O bioreactor include *Thauera*, *Gemmatimonadaceae* uncultured, *Nitrosomonadaceae* uncultured, *Ferruginibacter*, *Thiobacillus*, *Comamonadaceae* unclassified, *Betaproteobacteria* unclassified and *Blastocatella*. Among them, *Blastocatella* is especially highly enriched in the oxic tank. Five clusters were involved from cluster analysis.

The microbial community distribution in different tanks determines its functionality. The higher abundance of genera *Clostridiales* uncultured, *Christensenellaceae* uncultured (21%) belonging to class *Clostridium* was observed in MHA reactor. Bacteria affiliated to *Clostridium* are reported to be the acidogenic populations, which are able to utilize carbohydrates to generate H_2 and VFAs (Chu et al. 2012). The community *Clostridium* was reported to be predominant in dark fermentative H_2-producing bioreactor (Im et al. 2012). The members of genus *Gemmatimonadaceae* were detected in the activated sludge processes for sewage treatment under both anaerobic and aerobic conditions (Xu et al. 2017). The genus *Fastidiosipila* was identified in the anaerobic biomethanization consortium of municipal solid waste (Bertin et al. 2012). A strain of *Fastidiosipila* was report to grow under both anaerobic conditions and the presence of 2% and 6% O_2 (Falsen et al. 2005). The strains of *Anaerovorax* were found to be strictly anaerobic, which could ferment putrescine to acetate, butyrate, H_2 and ammonium (Matthies et al. 2000).

Some strains belonging to genus *Blastocateewalla* were aerobic chemoorganotrophic organisms and isolated from the soil (Foesel et al. 2013). The members of genus *Ferruginibacter* were reported to be Gram-negative, strictly aerobic rods and isolated from wetland freshwater (Kang et al. 2015). The higher abundance of *Nitrosomonadaceae* was consisted with the good nitrification in A/O reactor. The members of *Thauera* (Morgan-Sagastume et al. 2008), *Thiobacillus* (Li et al. 2016), *Comamonadaceae* and *Betaproteobacteria* (Wang and Chu 2016) were reported to have the function of denitrification. In addition, the bacteria having high abundance in the MHA and A/O system include glycogen-accumulating organisms *Defluviicoccus* for enhanced phosphorus removal (Burow et al. 2007), phenol-degrading bacterium *Ottowia* (Cao et al. 2014), syntrophic fatty acids-oxidizing bacteria *Anaerolineae* (Narihiro et al. 2012) and sulfur oxidizing bacterium *Limnobacter* (Chen et al. 2016).

It is clear that the sludge in the MHA reactor exhibited a plentiful community diversity. In addition to the dominant bacteria responsible for hydrolysis and acidification such as *Clostridiales*, *Anaerovorax*, *Anaerolineaceae* uncultured and *Fastidiosipila* (total abundance 35%), the aerobic nitrifiers such as *Nitrosomonadaceae* uncultured and *Nitrospira* were identified with a relatively high abundance (1.3% and 3.3% respectively). This might explain the increased nitrate concentration in MHA effluent (Fig. 3). The denitrifying bacteria *Thauera* was also found in MHA sludge with high abundance (6.4%). It is also noted that there were co-existed the sulfate reducing bacteria (SRB) such as *Desulfobacter*, *Desulfomicrobium* and *Desulfobulbus*, etc. (total abundance 4.2%) and the sulphur oxidizing bacteria (SOB) such as

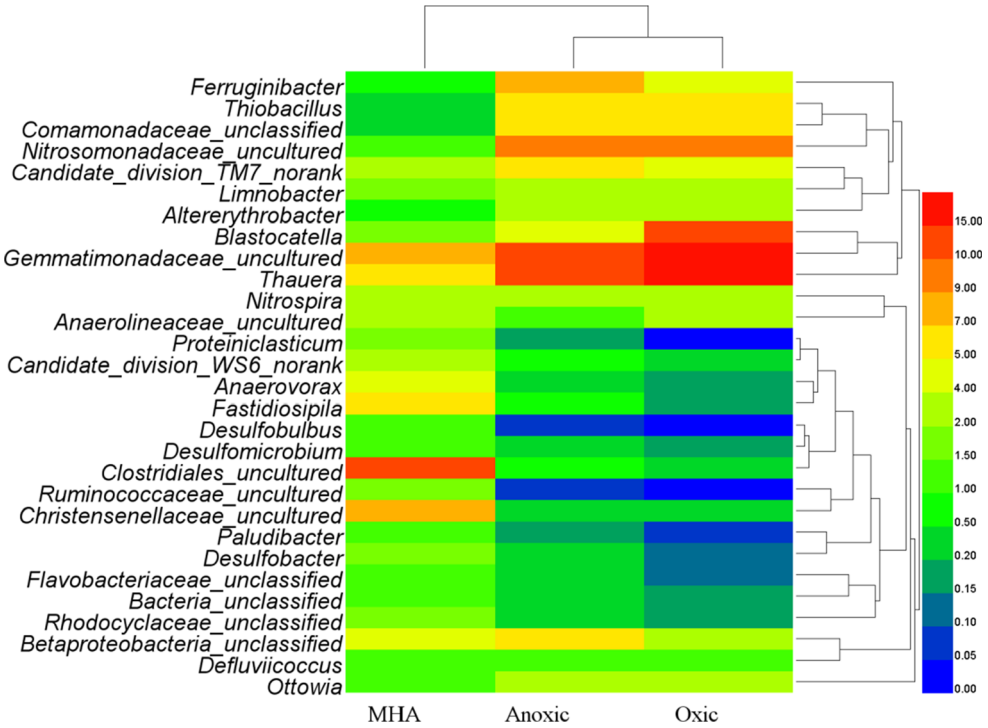


Fig. 8 The heatmap of genera of sludge samples in the three tanks (color bar representing the range of a genus and clustering were shown in the right side, abundance >0.10%)

Thiobacillus Arcobacter and *Limnobacter* (total abundance 2.8%) in the MHA bioreactor. Generally, SRB are considered to be obligate anaerobes. Recent studies found that SRB were able to survive in the presence of limited oxygen (Ontiveros-Valencia et al. 2014). The studies by Jenicek et al. (2011) demonstrated that the activity of SOB in the microaerobic sludge (10.4 mg/g h) was significantly higher in comparison with the anaerobic sludge (6.1 mg/g h), while the activity of SRB was not affected with limited air supply of 2.0 mg/g h. Wu et al. (2015) documented that SRB occupied for 2.8% in the lab-scale anaerobic HA reactor and their abundance dropped to 0.9% under microaerobic conditions.

Conclusions

The results of the pilot-scale study indicated that MHA pre-treatment coupled with A/O bioreactor was promising in treating petrochemical wastewater. MHA pre-treatment ensured the efficacy and stability of the following A/O treatment. Although the influent fluctuated, the efficiency of COD and NH_4-N removal maintained at around 78.3% and 80.8%, respectively. Nearly half of organic pollutants disappeared after MHA treatment, and only two compounds were detected and their abundance declined greatly after A/O treatment. The ratio of VFA/COD increased by 43–90% in the MHA effluent and the S^{2-} concentration (0.2 mg/L) was low with 5.5–13.8 L^{-} air m^{-3} h^{-1}.

The MHA sludge exhibited the higher protease activity, whereas the dehydrogenases activity was higher for the sludge in A/O tanks. Illumina MiSeq sequencing showed that the MHA sludge had the highest community richness and diversity and the predominant genera were *Clostridiales uncultured*, *Gemmatimonadaceae* uncultured, *Christensenellaceae* uncultured, *Thauera*, *Fastidiosipila*, *Betaproteobacteria* unclassified and *Anaerovorax*. In addition to the dominant bacteria responsible for hydrolysis and acidification, SRB, SOB, ammonium-oxidizing bacteria, nitrite-oxidizing bacteria and denitrifies were coexisted in MHA reactor.

Acknowledgements We would like to express sincere thanks to the technical staffs in the Petrochemical Wastewater Treatment Plant for the built-up and operation of the pilot-scale system. The research was financially supported by the National S&T Major Project (Grant No. 2012ZX07201-005-06-01).

Data and materials availability All data generated or analyzed during this study are included in this published article.

Author contribution Libing Chu was the major contributor in writing the original draft, reviewing and editing the manuscript. Pengyuan Ding performed the pilot-scale experiments. Mincong Ding contributed to the data analysis and visualization. All authors read and approved the final manuscript.

Funding We thank the financial support from the Key Program for Intergovernmental S&T Innovative Cooperation Project of China (No. 2017YFE0127000).

Declaration

Ethics approval and consent to participate Not applicable

Consent for publication Not applicable

Competing interests The authors declare no competing interests.

References

APHA/AWWA/AWEF, 2017. Standard Methods for the Examination of Water and Wastewater Washington, DC

Bertin L, Bettini C, Zanaroli G, Fraraccio S, Negroni A, Fava F (2012) Acclimation of an anaerobic consortium capable of effective biomethanization of mechanically-sorted organic fraction of municipal solid waste through a semi-continuous enrichment procedure. J. Chem. Technol. Biotechnol. 87:1312–1319

Burow LC, Kong Y, Nielsen JL, Blackall LL, Nielsen PH (2007) Abundance and ecophysiology of *Defluviicoccus* spp., glycogen-accumulating organisms in full-scale wastewater treatment processes. Microbiology 153:178–185

Cao J, Lai Q, Liu Y, Li G, Shao Z (2014) *Ottowia beijingensis* sp. nov., isolated from coking wastewater activated sludge, and emended description of the genus *Ottowia*. Int. J. Syst. Evol. Microbiol. 64: 963–967

Chen Q, Wu W, Qi D, Ding Y, Zhao Z (2020) Review on microaeration-based anaerobic digestion: state of the art, challenges, and prospectives. Sci. Total Environ. 710:136388

Chen S, Sun DZ, Chung JS (2008) Simultaneous removal of COD and ammonium from landfill leachate using an anaerobic-aerobic moving-bed biofilm reactor system. Waste Manage. 28:339–346

Chen Y, Feng X, He Y, Wang F (2016) Genome analysis of a *Limnobacter* sp. identified in an anaerobic methane-consuming cell consortium. Front. Mar. Sci. 3:1–8

Chu C-F, Ebie Y, Xu K-Q, Li Y-Y, Inamori Y (2012) Characterization of microbial community in the two-stage process for hydrogen and methane production from food waste. Int. J. Hydrogen Energy 35: 8253–8261

Diaz I, Perez SI, Ferrero EM, Fdz-Polanco M (2011) Effect of oxygen dosing point and mixing on the microaerobic removal of hydrogen sulphide in sludge digesters. Bioresour. Technol. 102:3768–3775

Ding P, Chu L, Wang J (2016) Biological treatment of actual petrochemical wastewater using anaerobic/anoxic/oxic process and the microbial diversity analysis. Appl. Microbiol. Biotechnol. 100:10193–10202

Diya'uddeen BH, Daud WMAW, Abdul Aziz AR (2011) Treatment technologies for petroleum refinery effluents: A review. Process Saf. Environ. 89:95–105

do Nascimento JGdS, de Araújo MHP, dos Santos AB, da Silva MER, Firmino PIM (2021) Redox mediator, microaeration, and nitrate addition as engineering approaches to enhance the biotransformation of antibiotics in anaerobic reactors. J. Hazard. Mater. 403: 123932.

Falsen E, Collins MD, Welinderolsson C, Song Y, Finegold SM, Lawson PA (2005) *Fastidiosipila sanguinis* gen. nov., sp. nov., a new Gram-positive, coccus-shaped organism from human blood. Int. J. Syst. Evol. Microbial. 55:853–858

Foesel BU, Rohde M, Overmann J (2013) *Blastocatella fastidiosa* gen. nov., sp. nov., isolated from semiarid savanna soil — the first described species of *Acidobacteria* subdivision 4. Syst. App. Microbiol. 36:82–89

Gessesse A, Dueholm T, Petersen SB, Nielsen PH (2003) Lipase and protease extraction from activated sludge. Water Res. 37:3652–3657

Guo C, Chen Y, Chen J, Wang X, Zhang G, Wang J, Cui W, Zhang Z (2014) Combined hydrolysis acidification and bio-contact oxidation system with air-lift tubes and activated carbon bioreactor for oilfield wastewater treatment. Bioresour. Technol. 169:630–636

Im WT, Kim DH, Kim KH, Kim MS (2012) Bacterial community analyses by pyrosequencing in dark fermentative H 2 -producing reactor using organic wastes as a feedstock. Inter. J. Hydrogen Energy 37: 8330–8337

Jenicek P, Celis CA, Koubova J, Pokorna D (2011) Comparison of microbial activity in anaerobic and microaerobic digesters. Water Sci. Technol. 63:2244–2249

Jenicek P, Keclik F, Maca J, Bindzar J (2008) Use of microaerobic conditions for the improvement of anaerobic digestion of solid wastes. Water Sci. Technol. 58:1491–1496

Kakavandi B, Babaei AA (2016) Heterogeneous Fenton-like oxidation of petrochemical wastewater using a magnetically separable catalyst (MNPs@ C): process optimization, reaction kinetics and degradation mechanisms. RSC Advances 6:84999–85011

Kang H, Kim H, Joung Y, Jang TY, Joh K (2015) *Ferruginibacter paludis* sp. nov., isolated from wetland freshwater, and emended descriptions of *Ferruginibacter lapsinanis* and *Ferruginibacter alkalilentus*. Int. J. Syst. Evol. Microbial. 65

Kim YK, Bae JH, Oh BK, Lee WH, Choi JW (2002) Enhancement of proteolytic enzyme activity excreted from Bacillus stearothermophilus for a thermophilic aerobic digestion process. Bioresource Technology 82:157–164

Krayzelova L, Bartacek J, Díaz I, Jeison D, Volcke EIP, Jenicek P, Jeison D (2015) Microaeration for hydrogen sulfide removal during anaerobic treatment: a review. Rev. Environ. Sci. Biotechnol. 14:703–725

Lazarova V, Manem J (1995) Biofilm characterization and activity analysis in water and wastewater treatment. Water Res. 29:2227–2245

Li R, Feng C, Hu W, Xi B, Chen N, Zhao B, Liu Y, Hao C, Pu J (2016) Woodchip-sulfur based heterotrophic and autotrophic denitrification (WSHAD) process for nitrate contaminated water remediation. Water Res. 89:171–179

Lim JW, Chiam JA, Wang J-Y (2014) Microbial community structure reveals how microaeration improves fermentation during anaerobic co-digestion of brown water and food waste. Bioresour. Technol. 171:132–138

Lin C-K, Tsai T-Y, Liu J-C, Chen M-C (2001) Enhanced biodegradation of petrochemical wastewater using ozonation and bac advanced treatment system. Water Res. 35:699–704

Loughrin J, Antle S, Bryant M, Berry Z, Lovanh N (2020) Evaluation of microaeration and sound to increase biogas production from poultry litter. Environments 7:62

Matthies C, Evers S, Ludwig W, Schink B (2000) *Anaerovorax odorimutans* gen. nov., sp. nov., a putrescine-fermenting, strictly anaerobic bacterium. Int. J. Syst. Evol. Microbial. 50:1591–1594

Morgan-Sagastume F, Nielsen JL, Nielsen PH (2008) Substrate-dependent denitrification of abundant probe-defined denitrifying bacteria in activated sludge. FEMS Microb. Ecol. 66:447–461

Nakasaki K, Tran LTH, Idemoto Y, Abe M, Rollon AP (2009) Comparison of organic matter degradation and microbial community during thermophilic composting of two different types of anaerobic sludge. Bioresour. Technol. 100:676–682

Narihiro T, Terada T, Ohashi A, Kamagata Y, Nakamura K, Sekiguchi Y (2012) Quantitative detection of previously characterized syntrophic bacteria in anaerobic wastewater treatment systems by sequence-specific rRNA cleavage method. Water Res. 46:2167–2175

Ontiveros-Valencia A, Tang Y, Krajmalnik-Brown R, Rittmann BE (2014) Managing the interactions between sulfate- and perchlorate-reducing bacteria when using hydrogen-fed biofilms to treat a groundwater with a high perchlorate concentration. Water Res. 55:215–224

Ramos I, Fdz-Polanco M (2014) Microaerobic control of biogas sulphide content during sewage sludge digestion by using biogas production and hydrogen sulphide concentration. Chem. Eng. J. 250:303–311

Ruan D, Zhou Z, Pang H, Yao J, Chen G, Qiu Z (2019) Enhancing methane production of anaerobic sludge digestion by microaeration: enzyme activity stimulation, semi-continuous reactor validation and microbial community analysis. Bioresour. Technol. 289:121643

Rubio-Clemente A, Chica E, Penuela GA (2015) Petrochemical wastewater treatment by photo-fenton process. Water Air Soil Pollut. 226: 62–62

Snaidr J, Amann R, Huber I, Ludwig W, Schleifer KH (1997) Phylogenetic analysis and in situ identification of bacteria in activated sludge. Appl. Environ. Microb. 63:2884–2896

Song G, Yu Y, Liu T, Xi H, Zhou Y (2019) Performance of microaeration hydrolytic acidification process in the pretreatment of 2-butenal manufacture wastewater. J. Hazard. Mater. 369:465–473

Tarjányi-Szikora S, Oláh J, Makó M, Palkó G, Barkács K, Záray G (2013) Comparison of different granular solids as biofilm carriers. Microchem. J. 107:101–107

Wang JL, Chu L (2016) Biological nitrate removal from water and wastewater by solid-phase denitrification process. Biotechnol. Adv. 34: 1103–1112

Wang K, Li W, Gong X, Li X, Liu W, He C, Wang Z, Minh QN, Chen C-L, Wang J-Y (2014) Biological pretreatment of tannery wastewater using a full-scale hydrolysis acidification system. Int. Biodeterior. Biodegrad. 95:41–45

Wu C, Li Y, Zhou Y, Li Z, Zhang S, Liu H (2018) Upgrading the Chinese biggest petrochemical wastewater treatment plant: technologies research and full scale application. Sci. Total Environ. 633:189–197

Wu C, Zhou Y, Sun Q, Fu L, Xi H, Yu Y, Yu R (2016) Appling hydrolysis acidification-anoxic–oxic process in the treatment of petrochemical wastewater: from bench scale reactor to full scale wastewater treatment plant. J. Hazard. Mater. 309:185–191

Wu C, Zhou Y, Wang P, Guo S (2015) Improving hydrolysis acidification by limited aeration in the pretreatment of petrochemical wastewater. Bioresour. technol. 194:256–262

Xu D, Liu S, Chen Q, Ni J (2017) Microbial community compositions in different functional zones of Carrousel oxidation ditch system for domestic wastewater treatment. AMB Express 7:40

Yang Q, Xiong P, Ding P, Chu L, Wang J (2015) Treatment of petrochemical wastewater by microaerobic hydrolysis and anoxic/oxic processes and analysis of bacterial diversity. Bioresour. Technol. 196:169–175

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.